

IMO Training – Number Theory

Hard Exercises

Longtin Chan
longtinclt@gmail.com

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Instructions:

- Write your solutions legibly on separate sheets of paper.
- If you wish your work to be marked, please scan your work and submit (in pdf format) through email.

1. (Hong Kong TST 2005)

Let r be the positive root of the equation $x^2 - 2004x - 1 = 0$. Define the sequence a_n by

$$a_0 = 1, a_{n+1} = [ra_n] \text{ for all } n \geq 0,$$

where $[y]$ denotes the greatest integer not exceeding y . Find the remainder when a_{2004} is divided by 2004.

2. (Hong Kong TST 2006) Find 2^{2006} positive integers satisfying the following conditions.

- (a) Each positive integer has 2^{2005} digits.
- (b) Each positive integer only has 7 or 8 in its digits.
- (c) Among any two chosen integers, at most half of their corresponding digits are the same.

3. (Hong Kong TST 2007)

Determine all ordered pairs (x, y) of positive integers such that $\frac{x^2y + x + y}{xy^2 + y + 11}$ are integers.

4. (Hong Kong TST 2009)

Show that the equation $y^{37} \equiv x^3 + 11 \pmod{p}$ is solvable for every prime p , where $p \leq 100$.

5. (Hong Kong TST 2011) Let p be a prime.

- (a) Show that there exists a prime divisor q of $\frac{p^p - 1}{p - 1}$ such that $q \not\equiv 1 \pmod{p}$.
- (b) Suppose q is a prime divisor of $\frac{p^p - 1}{p - 1}$ such that $q \not\equiv 1 \pmod{p}$. Show that $n^p - p \not\equiv 0 \pmod{q}$ for all positive integers n .

6. (Hong Kong TST 2012)

Let x, y, z, u, v , and w be integers satisfying $x^2 + y^2 = u^2$, $x^2 + z^2 = v^2$ and $y^2 + z^2 = w^2$. Find an integer l so that $518000 < l < 518518$ and $l \mid xyzuvw$.

7. (Hong Kong TST 2013)

Find a $m \in \mathbb{N}$ s.t. $x^3 \equiv 1 \pmod{m}$ has more than 2012 solutions.

(Note: If $x \equiv y \pmod{m}$, then x and y are considered as the same solution.)